

A COUNTEREXAMPLE TO THE KIMURA–MATSUMOTO–SATO ETERNAL DOMINATION CONJECTURE

ABSTRACT. Let $\gamma^\infty(G)$ be the eternal domination number of G in the model in which one guard moves per attack, and let $\theta(G)$ be the clique cover number. Conjecture 2.1 of Kimura, Matsumoto and Sato asserts that for a vertex-critical triangle-free graph G of order n with $|E(G)| \leq 2n - 3$, one has $\gamma^\infty(G) = \theta(G)$ if and only if $\gamma^\infty(G - v) = \theta(G - v)$ for every vertex v . We exhibit a single graph G_0 that refutes the “if” half. It has order 15 and exactly $27 = 2n - 3$ edges, so it lies on the conjecture’s own edge boundary; it is triangle-free and vertex-critical; $\gamma^\infty(G_0) = 7 < 8 = \theta(G_0)$; and $\gamma^\infty(G_0 - v) = \theta(G_0 - v) = 7$ for every one of its 15 vertices. The graph is the order-13 graph of the authors’ own Figure 2.1 with one odd ear of length three attached. The machine content is sixteen lower-bound tests: the fifteen bounds $\gamma^\infty(G_0 - v) > 6$, together with $\gamma^\infty(G_0) > 6$, which is what pins the value 7 rather than only $\gamma^\infty(G_0) \leq 7$. Each is certified by a well-founded attacker ranking over at most 1313 six-sets.

1. THE CONJECTURE

All graphs are finite, simple and undirected. In the *eternal domination* model of Goddard, Hedetniemi and Hedetniemi [1], a set $S \subseteq V(G)$ of guarded vertices must dominate G ; an attack at a vertex $r \notin S$ must be answered by moving one guard from some $w \in S$ with $wr \in E(G)$ to r ; and the resulting set must again dominate G and be able to answer any further attack, indefinitely. The least size of a set that can defend forever is the *eternal domination number* $\gamma^\infty(G)$. Writing $\alpha(G)$ for the independence number and $\theta(G)$ for the clique cover number, [1] gives

$$(1) \quad \alpha(G) \leq \gamma^\infty(G) \leq \theta(G).$$

Whether $\gamma^\infty = \theta$ can fail was raised by Klostermeyer and Mynhardt [3]; it can, and MacGillivray, Mynhardt and Virgile [5] exhibit triangle-free examples.

Kimura, Matsumoto and Sato [2] study the question inside vertex-critical graphs. Their Fact 2.1 records that $\theta(G) = \chi(\overline{G})$ and that G is *vertex-critical* exactly when $\overline{G}$ is vertex χ -critical, that is,

$$(2) \quad \theta(G - v) = \theta(G) - 1 \quad \text{for every } v \in V(G).$$

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Criticality is therefore a condition on θ , not on γ^∞ . Their Corollary 2.1 adds that a connected vertex-critical triangle-free graph has odd order n and $\theta = (n + 1)/2$. They then pose the following.

Conjecture 2.1 (Kimura, Matsumoto, Sato). Let G be a vertex-critical triangle-free graph of order n . If $|E(G)| \leq 2n - 3$, then

$$\gamma^\infty(G) = \theta(G) \iff \gamma^\infty(G - v) = \theta(G - v)$$

for any vertex v in $V(G)$.

The display above is transcribed from journal page 47 of [2], which is open access; the transcription, the numbering and the locator are bibliographic facts that the accompanying program does not check. What is refuted below is the displayed statement, with “for any vertex v ” read as ranging over all of $V(G)$.

The forward implication is the easy half, so a refutation must make the right-hand side true and the left-hand side false: we need a graph in the region whose every vertex-deleted subgraph is *tight*, while the graph itself is not.

2. THE WITNESS

Let H^* be the order-13 graph drawn as Figure 2.1 of [2], with graph6 string

L?`@?boNAs0wY0

which is also the fourth entry of Table 10 of [5]. It has 24 edges, is triangle-free and factor-critical, and satisfies $\theta(H^*) = 7$ and $\gamma^\infty(H^*) < 7$; the strict inequality is the content of the caption of [2, Figure 2.1], so

$$(3) \quad \gamma^\infty(H^*) \leq 6.$$

Define

$$G_0 = H^* + \text{the odd ear } 0\text{--}13, 13\text{--}14, 14\text{--}1,$$

a graph of order 15 whose 27 edges are, on the vertex set $\{0, 1, \dots, 14\}$,

$$\begin{array}{cccccccccc} 0\text{--}4 & 0\text{--}8 & 0\text{--}11 & 0\text{--}13 & 1\text{--}5 & 1\text{--}8 & 1\text{--}10 & 1\text{--}12 & 1\text{--}14 \\ 2\text{--}6 & 2\text{--}8 & 2\text{--}9 & 2\text{--}12 & 3\text{--}7 & 3\text{--}8 & 3\text{--}9 & 3\text{--}10 & 4\text{--}9 \\ 4\text{--}10 & 4\text{--}12 & 5\text{--}9 & 5\text{--}11 & 6\text{--}10 & 6\text{--}11 & 7\text{--}11 & 7\text{--}12 & 13\text{--}14 \end{array}$$

equivalently the graph6 string $N?`@?boNAs0wY0_?G?G$. Its degree sequence is $(4, 5, 4, 4, 4, 3, 3, 3, 4, 4, 4, 4, 4, 2, 2)$; deleting the two degree-2 vertices 13 and 14 returns H^* .

Theorem 1. *The graph G_0 is triangle-free and vertex-critical, of order $n = 15$ with $|E(G_0)| = 27 = 2n - 3$, and*

$$\gamma^\infty(G_0) = 7 < 8 = \theta(G_0) \quad \text{and} \quad \gamma^\infty(G_0 - v) = \theta(G_0 - v) = 7$$

for every one of the 15 vertices v . Consequently the “if” half of Conjecture 2.1 of [2] is false, and the conjecture is false as stated.

The graph G_0 is 2-connected.

3. PROOF

We use one lemma, which is not new.

Lemma 2 (subadditivity). *If $V(G) = V_1 \sqcup \cdots \sqcup V_t$, then*

$$\gamma^\infty(G) \leq \sum_{i=1}^t \gamma^\infty(G[V_i]).$$

Proof. For each i fix an eternal dominating family for $G[V_i]$ of size $\gamma^\infty(G[V_i])$ and let the guards on V_i play it, ignoring the other parts. A set that dominates V_i inside $G[V_i]$ dominates V_i inside G , so the union of the current sets dominates $V(G)$; and an attack at $r \in V_i$ is answered by a move inside V_i , which is legal in G because $E(G[V_i]) \subseteq E(G)$ and leaves every other part untouched. The union is therefore a valid configuration after every attack. $\square$

Special cases of Lemma 2 occur in the proofs of [2, Proposition 2.1] and [5, Corollary 5.1].

Proof of Theorem 1. Triangle-freeness. H^* is triangle-free, and the only new adjacencies are 0–13, 13–14 and 14–1. A triangle using them would need 0–14 or 1–13, and neither is an edge.

Size. $|E(G_0)| = 24 + 3 = 27 = 2 \cdot 15 - 3$.

Clique cover number. Since G_0 is triangle-free, every clique of G_0 has at most two vertices, so a clique cover is a matching together with the vertices it misses and $\theta(G_0) = 15 - \mu(G_0)$, where μ is the matching number. The matching

$$\{1-5, 2-8, 3-9, 4-10, 6-11, 7-12, 13-14\}$$

has 7 edges, and 7 cliques cover at most $14 < 15$ vertices, so $\mu(G_0) = 7$ and $\theta(G_0) = 8$. (The displayed matching plus the singleton $\{0\}$ is the corresponding cover of size 8.)

Vertex-criticality. By (2) and the previous paragraph, G_0 is vertex-critical if and only if $\theta(G_0 - v) = 7$, i.e. $\mu(G_0 - v) = 7$, for every v : that is, if and only if G_0 is factor-critical. Now H^* is factor-critical: it is connected, triangle-free and vertex-critical — [2, Figure 2.1] presents it as such — so $\mu(H^*) = 13 - \theta(H^*) = 6$ and, by the same equivalence applied to H^* , $\mu(H^* - v) = 6$ for every v , i.e. $H^* - v$ has a perfect matching. Attaching an odd ear preserves that; this is the ear step of Lovász’s odd ear decomposition [4], and we verify it directly. Let G be factor-critical, $a, b \in V(G)$, and let $G' = G + \{ax, xy, yb\}$ with $x, y \notin V(G)$, so $|V(G')|$ is again odd. For $v \in V(G)$, take a perfect matching M of $G - v$ and use $M \cup \{xy\}$ in $G' - v$; for $v = x$, take a perfect matching M of $G - b$ and use $M \cup \{yb\}$; for $v = y$, take a perfect matching M of $G - a$ and use $M \cup \{ax\}$. In each case $G' - v$ has a perfect matching, so G' is factor-critical. Hence G_0 is factor-critical, so vertex-critical, and $\theta(G_0 - v) = 7$ for all 15 vertices v .

The left-hand side fails. Partition $V(G_0) = V(H^*) \sqcup \{13, 14\}$. The second part induces a K_2 , and $\gamma^\infty(K_2) = 1$. By Lemma 2 and the published bound (3),

$$\gamma^\infty(G_0) \leq \gamma^\infty(H^*) + \gamma^\infty(K_2) \leq 6 + 1 = 7 < 8 = \theta(G_0),$$

so $\gamma^\infty(G_0) \neq \theta(G_0)$.

The right-hand side holds. Fix v . We have $\theta(G_0 - v) = 7$ and, by (1), $\gamma^\infty(G_0 - v) \leq 7$; so it suffices to show $\gamma^\infty(G_0 - v) > 6$. This is the machine step, and it is a finite emptiness test: $\gamma^\infty(H) \leq k$ holds if and only if the greatest fixed point of

$$\mathcal{F} \longmapsto \{S \in \mathcal{F} : \forall r \notin S \exists w \in S \cap N_H(r) \text{ with } S - w + r \in \mathcal{F}\},$$

started from the family of all dominating k -subsets of $V(H)$, is non-empty. For $H = G_0 - v$ and $k = 6$ there are at most $\binom{14}{6} = 3003$ candidate sets, and in each of the 15 cases the fixed point is empty. Together with $\gamma^\infty(G_0 - v) \leq 7 = \theta(G_0 - v)$ this gives $\gamma^\infty(G_0 - v) = \theta(G_0 - v) = 7$ for every v .

The right-hand side of Conjecture 2.1 therefore holds for G_0 and the left-hand side fails, while G_0 is vertex-critical, triangle-free and satisfies $|E(G_0)| \leq 2n - 3$. Finally $\gamma^\infty(G_0) = 7$ exactly, since $\gamma^\infty(G_0) \leq 7$ was shown above and the same emptiness test at $k = 6$ on G_0 itself (over $\binom{15}{6} = 5005$ candidates, of which 1313 are dominating) comes out empty. $\square$

4. SCOPE

We refute only the “if” half of the equivalence; nothing here bears on “ $\gamma^\infty(G) = \theta(G) \Rightarrow \gamma^\infty(G - v) = \theta(G - v)$ ”. Theorem 1 says the least order of a counterexample is at most 15; we do not claim it is 15, and no census of the smaller orders, or of order 15 itself, is carried out here. The value $\gamma^\infty(H^*) = 6$ is not published: what is in print is $\gamma^\infty(H^*) < \theta(H^*) = 7$, and (3) is all the proof uses.

5. REPRODUCTION AND VERIFICATION

Everything above can be redone from the printed edge list of G_0 . The machine step is sixteen emptiness tests on families of at most $\binom{15}{7} = 6435$ subsets, each recorded as a certificate.

In the positive direction the certificate is the surviving family itself, checked member by member to consist of dominating k -sets closed under every attack; for $\gamma^\infty(G_0) \leq 7$ it has 2196 members. In the negative direction it is a well-founded attacker ranking, so that no answer depends on an iteration having converged: every dominating k -set D carries the round $\text{rk}(D)$ at which it was discarded, and one checks that there is an attack vertex $r \notin D$ such that every legal response $D - w + r$ is either not dominating or has $\text{rk}(D - w + r) < \text{rk}(D)$; induction on rk then gives $\gamma^\infty(H) > k$. For $H = G_0 - v$ and $k = 6$ the numbers of dominating 6-sets

ranked, for $v = 0, 1, \dots, 14$ in turn, are

851, 721, 890, 890, 977, 1120, 1091, 1091,
1003, 890, 924, 757, 924, 1004, 1026,

and every one of them receives a rank. Neither check refers to the iteration that produced the data.

The accompanying program `verify.py` (Python 3.9 or later, standard library only, no external data file, no randomness, integer arithmetic throughout) is run by

`python3 verify.py`

and its recorded output is `verify.output.txt`. It reads the graph6 string and the edge list printed in Section 2, checks that they agree under the identity labelling, and re-derives the quantities stated above: the degree sequence; triangle-freeness; the maximum clique size 2; 2-connectedness; $\mu(G_0) = 7$ and $\theta(G_0) = 8$ with an explicit cover; factor-criticality, by computing a perfect matching of $G_0 - v$ for every v ; $\theta(G_0 - v) = 7$ and $\gamma^\infty(G_0 - v) = 7$ for all 15 vertices; $\gamma^\infty(G_0) = 7$; and the fifteen counts of dominating 6-sets displayed above. Every eternal-domination verdict is re-verified from its certificate in a separate pass. The program prints one line per check, exits 0 only if all pass, and states in its own output what it does not cover; the recorded run also contains checks of objects and cases beyond those the paper claims.

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